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Systems and methods for a multi-functional structural element of an inverter

Abstract

An inverter including: a housing including a first surface and a second surface opposite to the first surface; a power module including a first surface and a second surface opposite to the first surface, wherein the first surface of the power module contacts the first surface of the housing; a heat sink including a first surface and a second surface opposite to the first surface, wherein the first surface of the heat sink is coupled to the second surface of the power module; a multi-functional structural element, the multi-functional structural element including a first surface and a second surface, wherein the first surface of the multi-functional structural element contacts the second surface of the heat sink; and a printed circuit board, the printed circuit board being coupled to the second surface of the multi-functional structural element; wherein the housing and heat sink define a cooling circuit.

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Background/Summary

TECHNICAL FIELD

(1) Various embodiments of the present disclosure relate generally to an inverter systems, and, more particularly, to systems and methods for a multi-functional inverter structural element.

INTRODUCTION

(2) An inverter system may include a power module, inverter housing, and a printed circuit board

("PCB"). Thermal load management of an inverter system may be necessary to improve performance and reliability of the system. Cooling circuits may be used to transfer heat from the power module in order to cool the power module. Separately, thermal vias or passive cooling through an inverter housing may be used to cool the PCB. Improper cooling of the PCB or power module may lead to improper operations of the inverter system.

(3) The present disclosure is directed to overcoming one or more of these above-referenced challenges.

SUMMARY OF THE DISCLOSURE

(4) In some aspects, the techniques described herein related to an inverter including: a housing including a first surface and a second surface opposite to the first surface; a power module including a first surface and a second surface opposite to the first surface, wherein the first surface of the power module contacts the first surface of the housing; a heat sink including a first surface and a second surface opposite to the first surface, wherein the first surface of the heat sink is coupled to the second surface of the power module; a multi-functional structural element, the multi-functional structural element including a first surface and a second surface opposite to the first surface, wherein the first surface of the multi-functional structural element contacts the second surface of the heat sink; and a printed circuit board, the printed circuit board being coupled to the second surface of the multi-functional structural element; wherein the housing and heat sink define a cooling circuit, the cooling circuit including a first cooling channel and a second cooling channel, the first cooling channel being defined by a space in the housing and the second cooling channel being defined by inner walls of the heat sink.

(5) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is coupled to the heat sink by a thermal interface material.

(6) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is coupled to the heat sink by a hermetic seal.

(7) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is metal.

(8) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element does not directly contact the cooling circuit.

(9) In some aspects, the techniques described herein related to an inverter, wherein the heat sink is a diecast heat sink or sheet metal heat sink.

(10) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element includes a first plane, the first plane being a recess within the first surface of the multi-functional structural element, the recess being configured to receive the heat sink.

(11) In some aspects, the techniques described herein related to an inverter, wherein an air gap exists between the printed circuit board and an external cover of the housing.

(12) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element includes a fixation point capable of receiving a screw, the screw being configured to secure the printed circuit board and the multi-functional structural element to the housing.

(13) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element has a geometric cross section greater than the power module.

(14) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is configured to completely cover the printed circuit board from an electromagnetic field projected from the power module.

(15) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element includes a second plane that protrudes from the second surface of the multi-functional structural element and contacts the printed circuit board, wherein the second plane is configured to receive heat from the printed circuit board.

(16) In some aspects, the techniques described herein related to an inverter including: a housing including a first surface and a second surface opposite to the first surface; a power module including a first surface and a second surface opposite to the first surface, wherein the first surface of the power module contacts the first surface of the housing; a heat sink including a first surface and a second surface opposite to the first surface, wherein the first surface of the heat sink is coupled to the second surface of the power module; a multi-functional structural element, the multi-functional structural element including a first surface and a second surface opposite to the first surface, wherein the first surface of the multi-functional structural element contacts an outer wall of the heat sink, wherein the multi-functional structural element has a geometric cross section greater than the power module; and a printed circuit board, the printed circuit board being coupled to the second surface of the multi-functional structural element, wherein the multi-functional structural element is configured to completely cover the printed circuit board from an electromagnetic field projected from the power module; wherein the housing and heat sink define a cooling circuit, the cooling circuit including a first cooling channel and a second cooling channel, the first cooling channel being defined by a space in the housing and the second cooling channel being defined by inner walls of the heat sink.

(17) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is coupled to the heat sink by a thermal interface material.

(18) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is coupled to the heat sink by a hermetic seal.

(19) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element is metal.

(20) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element does not directly contact the cooling circuit.

(21) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element includes a first plane, the first plane being a recess within the first surface of the multi-functional structural element, the recess being configured to receive the heat sink.

(22) In some aspects, the techniques described herein related to an inverter, wherein the multi-functional structural element includes a second plane that protrudes from the second surface of the multi-functional structural element and contacts the printed circuit board, wherein the second plane is configured to receive heat from the printed circuit board.

(23) In some aspects, the techniques described herein related to a multi-functional structural element for an inverter, the element including: a first plane with a first surface and a second surface opposite to the first surface, the multi-functional structural element further including a second plane and a third plane, the second plane being a rectangular recess within the first surface of the first plane, the second plane being configured to receive and transfer heat to a heat sink, the third plane being a protrusion from the second surface of the first plane, the third plane being configured to contact a printed circuit board and configured to transfer heat from the printed circuit board to the first plane.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

(2) FIG. 1 depicts an exemplary system infrastructure for a vehicle including a combined inverter and converter including an inverter package system, according to one or more embodiments.

(3) FIG. 2 depicts an exemplary inverter package including a multi-functional inverter structural element, according to one or more embodiments.

(4) FIG. 2A depicts an exemplary power module and a multi-functional inverter structural element, according to one or more embodiments.

(5) FIG. 3 depicts an exemplary inverter package including a multi-functional inverter structural element connected to a heat sink by a thermal interface material, according to one or more embodiments.

(6) FIG. 4A depicts an exemplary inverter package including a multi-functional inverter structural element connected to a sheet metal heat sink by welding, according to one or more embodiments.

(7) FIG. 4B depicts an exemplary inverter package including a multi-functional inverter structural element connected to a diecast heat sink by a seal and screws, according to one or more embodiments.

(8) FIG. 4C depicts an exemplary inverter package including a multi-functional inverter structural element connected to a sheet metal heat sink by a thermal interface material, according to one or more embodiments.

(9) FIG. 5 depicts an exemplary multi-functional inverter structural element in a housing, according to one or more embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. In this disclosure, unless stated otherwise, relative terms, such as, for example, “about,” “substantially,” and “approximately” are used to indicate a possible variation of +10% in the stated value. In this disclosure, unless stated otherwise, any numeric value may include a possible variation of +10% in the stated value.

(11) The terminology used below may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. For example, in the context of the disclosure, the switching devices may be described as switches or devices, but may refer to any device for controlling the flow of power in an electrical circuit. For example, switches may be metal-oxide-semiconductor field-effect transistors (MOSFETs), bipolar junction transistors (BJTs), insulated-gate bipolar transistors (IGBTs), or relays, for example, or any combination thereof, but are not limited thereto.

(12) Various embodiments of the present disclosure relate generally to an inverter system, and more particularly to, to systems and methods for a multi-functional inverter structural element.

(13) Inverters, such as those used to drive a motor in an electric vehicle, for example, are responsible for converting High Voltage Direct Current (“HVDC”) into Alternating Current (“AC”) to drive a motor. Inverters may include a housing, a power module, a printed circuit board (“PCB”), and a cooling circuit. Power module may include one or more silicon carbide (“SiC”)-based power switches that deliver relatively high power densities and efficiencies needed to extent battery range and performance. The power module may contain circuitry and components that are configured to convert DC current from the electric vehicle battery to AC current, which can be utilized within the electric motor that drives the propulsion system. The housing of an inverter may structurally connect the components of the inverter together. The PCB may include control and gate driver boards. The control and gate driver boards may further include active components such as power

supplies, active discharge, and gate drivers. The cooling circuit of the inverter may transfer waste heat (e.g., heat generated while in operation) from the system to a coolant associated with the fluid circuits and transfer the fluid to a radiator, heat exchanger, or other engine components, as described herein.

(14) Inverter systems may have high ambient temperature during operation. Cooling inverter systems may help improved performance and reliability. Some inverter systems utilize conventional thermal management methods to cool the active components of the PCB. These methods may include cooling the PCB through thermal vias in a copper plane connected to the PCB or by passive cooling through the housing of the inverter. Cooling the PCB through thermal vias may not provide significant cooling and may require oversizing of PCB components. Passive cooling of the PCB by the inverter housing may increase the complexity and cost of assembling inverter housing components (e.g., the inverter cover). Some inverter cooling systems may include individual cooling elements for the various components of a PCB. Some inverter systems may include power modules with double sided cooling performed by two coolant channel covers. The double coolant channels may offer no secondary benefits. Some inverter systems may have no active cooling for PCB components.

(15) Some inverter systems may mount a PCB to inverter housing by using screws. The screw mounts may cause structural inverter integrity concerns. Some inverter systems may include PCB fixation points connected directly to an inverter housing.

(16) Some inverter systems may include an EMC shield to protect noise coupling from an AC switching side of the power module and the PCB.

(17) One or more embodiments of the system described herein may provide thermal management of the power module with multiple cooling elements, thermal management of active components of the control and gate driver board, and thermal management of the inner temperature of the inverter.

(18) One or more embodiments may provide an inverter system that includes flexible mounting positions of the PCB. The flexible mount positions of the PCB may be capable of meeting different mechanical shock and vibration system requirements.

(19) One or more embodiments may provide an inverter structural element within the inverter system capable of Electromagnetic (EMC) shielding of PCB components such as the control and gate driver board.

(20) One or more embodiments may the decrease the complexity and cost of assembling inverter housing components (e.g., an inverter cover).

(21) One or more embodiments may include a multi-functional structural element (“MFSE”) within an inverter system. As will be described in greater detail below, the MFSE may be configured to dissipate heat from the active components (e.g., power supplies, active discharge, gate drivers) from the control and gate driver boards of the PCB. This active cooling of the PCB may lead to improved reliability and allow for more cost-effective sizing of PCB components. The MFSE may be configured to enable heat transfer from a first side in contact with a PCB and a second side in contact with a heat sink including a cooling channel. The MFSE may be configured to reduce the inner inverter temperature through the heat transfer capabilities. The MFSE may be utilized in combination with double side cooled power switches to enable efficient cooling of the inverter components while easing the assembly process. For example, one of the coolant channels utilized to cool the power switches may be used, in combination with the MFSE to cool the PCB.

(22) The MFSE may be utilized as a structural element within an inverter system and be configured to fixate to the PCB. The fixation may allow for fixation points to be placed at optimal positions to reduce mechanical PCB stress and minimize scrappage. For example, fixation points may be located at the four corners of the MFSE as well as at additional locations such as adjacent to raised planes of the MFSE. Some fixation points may allow for fixation of the MFSE to the housing and PCB. Other fixation points may only fixate the MFSE to the PCB. The MFSE may be a separate mountable element that may be mounted to a coolant channel with a thermal interface material

("TIM"), by a hematic seal, or by a seal and screws.

(23) The MFSE may be configured to provide EMC shielding for the control and gate driver boards of the PCB from the power module of the inverter.

(24) FIG. 1 depicts an exemplary system infrastructure for a vehicle including a combined inverter and converter including an inverter package system, according to one or more embodiments. In the context of this disclosure, the combined inverter and converter may be referred to as an inverter. As shown in FIG. 1, electric vehicle **100** may include an inverter **110**, a motor **190**, and a battery **195**. The inverter **110** may include components to receive electrical power from an external source and output electrical power to charge battery **195** of electric vehicle **100**. The inverter **110** may include an inverter package **200**. The inverter **110** may convert DC power from battery **195** in electric vehicle **100** to AC power, to drive motor **190** of the electric vehicle **100**, for example, but the embodiments are not limited thereto. The inverter **110** may be bidirectional, and may convert DC power to AC power, or convert AC power to DC power, such as during regenerative braking, for example. Inverter **110** may be a three-phase inverter, a single-phase inverter, or a multi-phase inverter.

(25) FIG. 2 depicts an exemplary inverter package **200** including a multi-functional inverter structural element ("MFSE") **202**, according to one or more embodiments. The inverter package **200** may include an inverter housing **204**, a MFSE **202**, a printed circuit board ("PCB") **206**, and a power module **208** (e.g., one or more power module, e.g., three power modules, e.g., six power modules). The inverter package **200** may include a first cover **210**, a second cover **212**, a cooling circuit **214**, and an open space **216**. The inverter housing may be configured to secure the power module **208**, the MFSE **202**, the cooling circuit **214**, and the PCB **206**. The first cover **210** may be located at a first top end of the inverter package **200** and the second cover **212** may be located at a second bottom end opposite of the first cover **210**. The open space **216** (e.g., gap) may be located within the inverter housing **204** between the second cover **212** and the PCB **206**. The open space **216** may include elements mounted to the PCB **206**.

(26) The cooling circuit **214** may include an inlet **218**, an outlet **220**, a first side cooling channel **222**, and a second side cooling channel **224**. The cooling circuit **214** may transfer waste heat (e.g., heat generated while in operation) from the inverter package **200** to a coolant flowing through the cooling circuit **214** and transfer the fluid to a radiator, heat exchanger, or other engine components. The flow of the coolant may be indicated by the arrows in the cooling circuit **214**. The inlet **218** may be located at a third end of the inverter package **200**, perpendicular to the first and second end. The outlet **220** may be located at a fourth end of the inverter package **200**, also perpendicular to the first and second end, and located on the opposite side of the third end. The inlet **218** may be configured to receive a coolant into the cooling circuit **214** and the outlet **220** may expel the coolant from the cooling circuit **214**. The first side cooling channel **222** may be defined by a hollow opening within the inverter housing **204** and the first cover **210** of inverter housing **204**. Thus, a wall defining the hollow opening in the inverter housing **204** and a portion **250** of the surface of the first cover **210** may define the first side cooling channel **222**. The connection of the portion **250** and the section of the inverter housing **204** defining the first side cooling channel **222** may be a hermetic seal. A hermetic seal may be a stir friction welding seal, an ultra-sound welding seal, a soldering seal, or a gasket seal.

(27) The second side cooling channel **224** may be defined by a heat sink **228** having a hollow interior. The hollow interior of heat sink **228** may be configured to allow coolant to flow from a first end of heat sink **228** across the YC axis to a second end of heat sink **228**. The heat sink **228** may have a first surface **252** and a second surface **254** opposite of the first surface **252**. The first surface **252** of heat sink **228** may be in contact with the power modules **208**. The first surface **252** of heat sink **228** may be coupled to the power modules **208** by a thermal interface material. The heat sink **228** may be a sheet metal heat sink **428A** (as depicted in FIGS. 4A and 4C) or a diecast heat sink **428B** (as depicted in FIG. 4B).

(28) Portions of the inverter housing **204** may transfer heat to a coolant within the first side cooling channel **222**. The power module **208** along with portions of the inverter housing **204** may transfer heat to a coolant within the second side cooling channel **224** through the heat sink **228**.

(29) The second surface of the heat sink **228** may be connected to the first surface of the power module **208** and to the MFSE **202** by a hermetic seal.

(30) The MFSE **202** may include a first surface **239** and a second surface **240**, the first surface **239** being opposite to the second surface **240**. A first plane **544** (shown in FIG. 5) of the first surface **239** of the MFSE **202** may contact and receive the heat sink **228**. The first plane **544** may be a planar section that is a recess in the first surface **239**. The recess may be along the ZC axis. The heat sink **228** and first plane **544** connection may include a hermetic seal. The first plane **544** may be generally rectangular and indented to match and receive the heat sink **228**. The first plane **544** (shown in FIG. 5) may be capable of receiving heat from the rest of the MFSE **202** and transferring the heat to the heat sink **228** and then to the second side cooling channel **224**. The MFSE **202** in combination with the first side cooling channel **222** and second side cooling channel **224** may enable efficiently cooling over the inverter components.

(31) The first surface **239** of the MFSE **202** may contact portions of inverter housing **204** at a fixation point **236** (e.g., one or more fixation points, e.g., two fixations points, e.g., at least three fixation points). Fixation point **236** may extend through the first surface **239** to the second surface **240** of the MFSE **202**. The fixation point **236** may include a threaded opening configured to allow for a screw to connect the MFSE **202** and PCB **206** to the inverter housing **204**. This fixation point **236** may reduce mechanical PCB **206** stress. For example, the fixation point **236** may increase support for the PCB **206**, lowering the mechanical stress. The MFSE **202** may be made of a more rigid material in comparison to the PCB **206**. The fixation point **236** coupling the PCB **206** to the MFSE **202** may decrease the bending of the PCB **206** during use of the inverter package **200**.

(32) The second surface **240** of the MFSE **202** may contact and connect to a first surface of the PCB **206** at a fixation point **236**. The fixation point **236** may be located at optimal position so as to reduce the mechanical PCB **206** stress and minimize scrappage during use of the inverter package **200**. For example, there may be a fixation point **236** at each of the four corners of the MFSE **202**. The fixation point **236** may allow for a screw to extend from the PCB **206**, through the MFSE **202** and screw into the housing **204** (shown in FIG. 5). Additional fixation points **546** (shown in FIG. 5) may be located adjacent to each second plane **238**. The additional fixation points **546** may couple the MFSE **202** to the PCB **206**, but not couple to the housing **204**. In some embodiments, the PCB **206** may only be in contact the MFSE **202** and the PCB **206** may not directly contact the inverter housing **204** or the second cover **212**. The second surface **240** of the MFSE **202** may further contact the PCB **206** at a second plane **238** (e.g., one or more second planes, e.g., two second planes). The second plane **238** may protrude from the second surface of the MFSE **202** in the ZC axis. The second plane **238** may extend further in the ZC axis than the first plane **544**. The second plane **238** may be capable of receiving heat from the PCB **206** and transferring the heat through the MFSE **202** to the heat sink **228**. For example, this may performed by transfer of heat from the PCB **206**, through second plane **238**, to a first plane **544** (depicted in FIG. 5), through the first plane **544** (in the ZC axis direction) to the heat sink **228**, to the second side cooling channel **224**.

(33) An open space **216** may be located between a second surface of the PCB **206** and the second cover **212**, the second surface of the PCB **206** being opposite of the first surface of the PCB **206**. The open space **216** may not include air.

(34) The MFSE **202** may be made of metal such as copper, aluminum, different types of alloy, sheet metal parts, or a diecast part. The MFSE may configured to provide shielding of the PCB **206** from the power module **208**. For example, the material of the MFSE **202** may not allow for electromagnetic fields to pass through the MFSE **202**. FIG. 2A depicts an exemplary power module and a multi-functional inverter structural element, according to one or more embodiments. FIG. 2A may display the geometric relationship of the power module **208** and the MFSE **202**.

(35) For example, the cross section of the MFSE **202** in the XC, YC plane may be greater than the cross section of the power module **208**. The geometric position of the MFSE **202** combined with the greater gross sectional area may completely cover the ZC axis electromagnetic projection from the power module **208** as displayed in FIG. 2A. For example, the gate driver board and control boards of the PCB **206** may not be exposed to the electromagnetic fields from components of the power module **208**.

(36) FIG. 3 depicts an exemplary inverter package **300** including a multi-functional inverter structural element (MFSE) **202** connected to a heat sink **228** by a thermal interface material (TIM) **326**, according to one or more embodiments. TIM **326** may be located between the second surface **254** of the heat sink **228** and the MFSE **202**. The TIM **326** may include epoxy, silicon, or other elastomer products, for example. The TIM **326** may be provided as a dielectric pad, heat spreader, thermal compound, thermal gel (e.g., dispensable gel), thermal tape, phase change material, gap pad, thermal grease, or alternative thermal material, for example. An advantage of the inverter package **300** with TIM **326** is that the MFSE may be a stronger and/or more durable connection. The elements described above in FIG. 3 may be similar to corresponding elements described in FIG. 2 in structure and function.

(37) FIG. 4A depicts an exemplary inverter package **400A** including a multi-functional inverter structural element (“MFSE”) **202** connected to a sheet metal heat sink **428A** by welding, according to one or more embodiments. The sheet metal heat sink **428A** may define the second side cooling channel **224**. The sheet metal heat sink **428A** may be connected to the power module **208** at a first surface, the first surface being located closest to the power module **208**. A second surface of the sheet metal heat sink **428A** located closest to the PCB **206** may be received by the MFSE **202**. The elements described above in FIG. 4A may be similar to corresponding elements described in FIG. 2 in structure and function. The sheet metal heat sink **428A** may have a greater performance (e.g., thermal conductivity, and/or durability) than the diecast heat sink **428B** depicted in FIG. 4B. The inverter package **400A** with the sheet metal heat sink **428A** may have a greater thermal conductivity efficiency by utilizing a welding connecting between the MFSE **202** and the sheet metal heat sink **428A** in comparison to utilizing TIM **326** for the connection (e.g., as utilized in FIG. 4C).

(38) FIG. 4B depicts an exemplary inverter package **400B** including a multi-functional inverter structural element (“MFSE”) **202** connected to a diecast heat sink **428B** by a seal and screws, according to one or more embodiments. The connection between the MFSE **202** and diecast heat sink **428B** may be located at fixation point **542** (shown in FIG. 5) (e.g., one or more fixation points **542**, e.g., eight fixation points **542**). The diecast heat sink **428B** may define the second side cooling channel **224**. The diecast heat sink **428B** may be connected to the power module **208** at a first surface, the first surface being located closest to the power module **208**. This coupling may be performed by one or more types of TIM materials (e.g., soft TIM such as thermal paste or grease, e.g., hard TIM such as soldering or sintering). Opposite of the first surface, may be a second surface that includes an opening. The second surface of the diecast heat sink **428B** on the side located closest to the PCB **206** may be received and coupled to the MFSE **202**. This coupling may occur at the edge of the first plane **544** (not shown). The connection may include a seal **432** that is received within the second surface of the diecast heat sink **428B**. In inverter package **400B**, a portion of the MFSE **202** (e.g., the first plane **544** shown in FIG. 5) may define a portion (e.g., a face) of the second side-cooling channel **224**. elements described above in FIG. 4B may be similar to corresponding elements described in FIG. 2 in structure and function.

(39) FIG. 4C depicts an exemplary inverter package **300** including a multi-functional inverter structural element (“MFSE”) **202** connected to a sheet metal heat sink **428A** by a thermal interface material (“TIM”) **326**, according to one or more embodiments. The sheet metal heat sink **428A** may be connected to the power module **208** at a first surface, the first surface being located closest to the power module **208**, by a TIM **326**. The TIM **326** may be soft TIM such as thermal paste or

grease or a hard TIM such as soldering or sintering. A second surface of the sheet metal heat sink **428A** located closest to the PCB **206** may be received by the MFSE **202**. The elements described above in FIG. **4C** may be similar to corresponding elements described in FIG. **2** in structure and function.

(40) FIG. **5** depicts an exemplary multi-functional inverter structural element (“MFSE”) **202** in an inverter housing **204**, according to one or more embodiments. The MFSE **202** may contain a first surface (not shown in FIG. **5**) and a second surface **240** opposite to the first surface in the ZC axis. The MFSE **202** may include fixation points **542**, fixation points **236**, fixation points **546**, a first plane **544**, and a second plane **238** (e.g., two second planes). The second side cooling channel **224** may be located in a heat sink **228** (not shown) connected to the MFSE **202** at the first surface (not shown). The connection may be performed using a fixation point **542**. The fixation point **542** may include a thread configured to receive a screw to securely fix the heat sink **228** to the MFSE **202**. For example, the MFSE **202** may include eight fixation points **542**. Two fixation points **542** may be located adjacent to each side of the first plane **544**. The connection of the MFSE **202** and heat sink **228** may further be performed by a hermetic seal between the MFSE **202** and heat sink **228**. The connection of the MFSE **202** and heat sink **228** may further be performed by TIM **326** (not shown). (41) As discussed above, the second surface **240** of the MFSE **202** may be configured to receive the PCB **206**. The second surface **240** may be secured to the PCB **206** at fixation point **236** and fixation points **546**. For example, the MFSE **202** may include a fixation point **236** located at each corners of the XC-YX plane along the second surface **240** of the MFSE **202**. Fixation point **236** may include a thread configured to receive a screw to securely fix the MFSE **202** to the PCB **206** and inverter housing **204**. MFSE may include a fixation point **546** located next to each second plane **238**. The fixation point **546** may include a thread configured to receive a screw to securely fix the MFSE **202** to the PCB **206**.

(42) The MFSE **202** may further including first plane **544**. The first plane **544** may transfer heat from the MFSE **202** to the heat sink **228**.

(43) The second surface **240** of the MFSE **202** may further contact the PCB **206** at a second plane **238**. The second plane **238** may protrude from the second surface of the MFSE **202**. For example, the second plane **238** may include two second planes **238**. The second plane **238** may be capable of receiving heat from the PCB **206** (not shown) and transferring the heat through the MFSE **202** to the heat sink **228**. The second planes **238** may be positioned so as to contact elements from the PCB **206** that require extra cooling. For example, this may performed by transfer of heat from the PCB **206** (not shown), through second plane **238**, to a first plane **544** through the first plane **544** (in the ZC axis direction) to the heat sink **228**, to the second side cooling channel **224**. In some examples, the coolant does not directly contact the MFSE **202**. In other examples (as shown in FIG. **4B**), the coolant may contact the MFSE **202**. In some examples, the coolant may contact only the inner surface of inlet **218**, the inner surface of inverter housing **204**, the inner surface of heat sink **228**, the portion **250** of the first cover **210**, and the inner surface of outlet **220**.

(44) FIG. **2** depicts an exemplary inverter package **200** cooling circuit **214**, according to one or more embodiments. The arrows located within the cooling circuit **214** may indicate the flow of coolant through the cooling circuit **214**. Coolant may enter the cooling circuit through inlet **218**. Coolant may proceed to flow through the first side cooling channel **222** and through the second side cooling channel **224**, following the flow path of the arrows. The coolant may flow from inlet **218**, in the YC axis direction across the first side cooling channel **222** and out through the outlet **220**. Coolant may also, after entering through inlet **218** and entering the first side cooling channel **222**, flow substantially perpendicular to the first side cooling channel **222** in the ZC axis to enter the second side cooling channel **224**. Coolant may then flow in the YC axis direction across the second side cooling channel **224** and then flow substantially perpendicular to the second side cooling channel **224** in the YC axis and merge with the first side cooling channel **222**. As coolant flows through the cooling circuit, heat may flow from the inverter housing **204**, power module **208**,

PCB 206, and heat sink 228 (not shown) into the coolant. The heated coolant may then be expelled from the cooling circuit 214 through outlet 220.

(45) One or more embodiments may include an MFSE configured to dissipate heat from the active components (e.g., power supplies, active discharge, gate drivers) from the control and gate driver boards of the PCB. One or more embodiments may include an MFSE to provide structural support within an inverter system and allow for a secure connection of the PCB within the inverter system. One or more embodiments may include an MFSE configured to provide electromagnetic shielding of PCB components from the power module.

(46) Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. An inverter comprising: a housing including a first surface and a second surface opposite to the first surface; a power module including a first surface and a second surface opposite to the first surface, wherein the first surface of the power module contacts the first surface of the housing; a heat sink including a first surface and a second surface opposite to the first surface, wherein the first surface of the heat sink is coupled to the second surface of the power module; a multi-functional structural element, the multi-functional structural element including a first surface and a second surface opposite to the first surface, wherein the first surface of the multi-functional structural element contacts the second surface of the heat sink; and a printed circuit board, the printed circuit board being coupled to the second surface of the multi-functional structural element; wherein the housing and heat sink define a cooling circuit, the cooling circuit including a first cooling channel and a second cooling channel, the first cooling channel being defined by a space in the housing and the second cooling channel being defined by inner walls of the heat sink.
2. The inverter of claim 1, wherein the multi-functional structural element is coupled to the heat sink by a thermal interface material.
3. The inverter of claim 1, wherein the multi-functional structural element is coupled to the heat sink by a hermetic seal.
4. The inverter of claim 1, wherein the multi-functional structural element is metal.
5. The inverter of claim 1, wherein the multi-functional structural element does not directly contact the cooling circuit.
6. The inverter of claim 1, wherein the heat sink is a diecast heat sink or sheet metal heat sink.
7. The inverter of claim 1, wherein the multi-functional structural element includes a first plane, the first plane being a recess within the first surface of the multi-functional structural element, the recess being configured to receive the heat sink.
8. The inverter of claim 1, wherein an air gap exists between the printed circuit board and an external cover of the housing.
9. The inverter of claim 1, wherein the multi-functional structural element includes a fixation point capable of receiving a screw, the screw being configured to secure the printed circuit board and the multi-functional structural element to the housing.
10. The inverter of claim 1, wherein the multi-functional structural element has a geometric cross section greater than the power module.
11. The inverter of claim 10, wherein the multi-functional structural element is configured to completely cover the printed circuit board from an electromagnetic field projected from the power module.
12. The inverter of claim 1, wherein the multi-functional structural element includes a second plane that protrudes from the second surface of the multi-functional structural element and contacts the

printed circuit board, wherein the second plane is configured to receive heat from the printed circuit board.

13. An inverter comprising: a housing including a first surface and a second surface opposite to the first surface; a power module including a first surface and a second surface opposite to the first surface, wherein the first surface of the power module contacts the first surface of the housing; a heat sink including a first surface and a second surface opposite to the first surface, wherein the first surface of the heat sink is coupled to the second surface of the power module; a multi-functional structural element, the multi-functional structural element including a first surface and a second surface opposite to the first surface, wherein the first surface of the multi-functional structural element contacts an outer wall of the heat sink, wherein the multi-functional structural element has a geometric cross section greater than the power module; and a printed circuit board, the printed circuit board being coupled to the second surface of the multi-functional structural element, wherein the multi-functional structural element is configured to completely cover the printed circuit board from an electromagnetic field projected from the power module; wherein the housing and the heat sink define a cooling circuit, the cooling circuit including a first cooling channel and a second cooling channel, the first cooling channel being defined by a space in the housing and the second cooling channel being defined by inner walls of the heat sink.

14. The inverter of claim 13, wherein the multi-functional structural element is coupled to the heat sink by a thermal interface material.

15. The inverter of claim 13, wherein the multi-functional structural element is coupled to the heat sink by a hermetic seal.

16. The inverter of claim 13, wherein the multi-functional structural element is metal.

17. The inverter of claim 13, wherein the multi-functional structural element does not directly contact the cooling circuit.

18. The inverter of claim 13, wherein the multi-functional structural element includes a first plane, the first plane being a recess within the first surface of the multi-functional structural element, the recess being configured to receive the heat sink.

19. The inverter of claim 13, wherein the multi-functional structural element includes a second plane that protrudes from the second surface of the multi-functional structural element and contacts the printed circuit board, wherein the second plane is configured to receive heat from the printed circuit board.

20. A multi-functional structural element for an inverter, the element comprising: a first plane with a first surface and a second surface opposite to the first surface, the multi-functional structural element further including a second plane and a third plane, the second plane being a rectangular recess within the first surface of the first plane, the second plane being configured to receive and transfer heat to a heat sink, the third plane being a protrusion from the second surface of the first plane, the third plane being configured to contact a printed circuit board and configured to transfer heat from the printed circuit board to the first plane.
